Deep Learning Notes

1 Introduction to Deep Learning

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers.

2 Artificial Neural Network (ANN)

A neuron computes:

$$z = w^T x + b \tag{1}$$

$$a = f(z) \tag{2}$$

Where:

- w = weights
- x = input vector
- b = bias
- f = activation function

3 Activation Functions

3.1 Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{3}$$

3.2 ReLU

$$f(x) = \max(0, x) \tag{4}$$

3.3 Tanh

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{5}$$

4 Loss Functions

4.1 Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{6}$$

4.2 Binary Cross Entropy

$$L = -\left[y\log(\hat{y}) + (1-y)\log(1-\hat{y})\right] \tag{7}$$

5 Gradient Descent

Update rule:

$$w = w - \eta\frac{\partial L}{\partial w} \tag{8}$$

Where η is the learning rate.

6 Backpropagation

Chain rule:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial z}\cdot\frac{\partial z}{\partial w} \tag{9}$$

7 Recurrent Neural Network (RNN)

Hidden state:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \tag{10}$$

Output:

$$y_t = W_y h_t \tag{11}$$

8 Convolutional Neural Network (CNN)

Convolution operation:

$$S(i,j) = (I * K)(i,j) \tag{12}$$

9 Softmax Function

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{13}$$

10 Dropout

Randomly deactivate neurons during training to prevent overfitting.